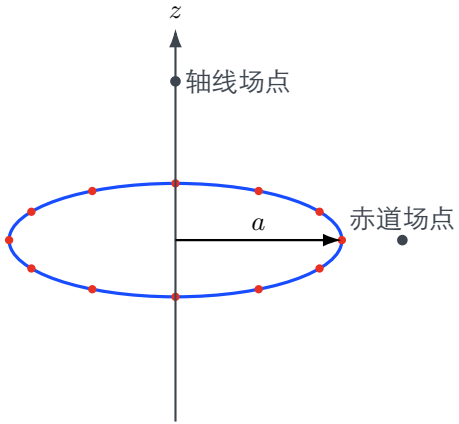
均匀带电圆环：轴线与赤道面的四极修正符号相反



$$\varphi = \frac{kQ}{r} - \frac{kQa^2}{2r^3}P_2(\cos\theta) + O(r^{-5})$$

$$\varphi_{\text{轴}} = \frac{kQ}{\sqrt{r^2+a^2}} = \frac{kQ}{r}\left(1 - \frac{a^2}{2r^2} + \cdots\right)$$

四极修正：轴线为负，赤道面为正。